

CSS385: Introduction to Game Development

M/W 5:45pm-7:45pm

Remote Learning: Please refer to [this page for Zoom URL](#)

COVID-19 Related Resources

Student Help for Learning Online

<https://www.uwb.edu/it/student-continuity>

Some [details on Zoom and remote learning](#)

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Email: ksung@uw.edu	Office Hours: Mon 12-2pm (or email for appointment)

Course Description

This class examines the fundamental issues in designing and developing computer video games. We begin with examining technical issues in developing a game including: software architecture, interaction model, mathematics concepts, real-time approximation for Newtonian physics, elementary special effects, and algorithms relevant to supporting in-game logic. After the technical knowledge, in the second half of the quarter we examine the elements in games design including: world setting, game play, and interface; and the procedure of designing games including: conceptualization, prototyping, and playtesting.

The first half of the class is programming intensive where students practice and implement the technical knowledge learned by developing games individually. After the mid-quarter, students will work in groups to design and implement their own games. The second half of the class involves group work, presentations, and active class participation in games evaluation.

Prerequisites: 2.0 in CSS 342 or equivalent.

Grading

Programming Assignments (4)	35%
Weekly Assignments + Classroom Participation	10%
Final Game	55%

Books

Required Textbook: *None*. Reference materials will be provided.

Reference Textbooks: (Posted off [Files/Reference Readings](#))

Introduction to game development, S. Rabin (Ed.), Boston, MA, Charles River Media, 2005.

- **A1:** Bakie, R. T. (2005). A brief history of video games. In S. Rabin (Ed.), *Introduction to game development* (pp. 1-36). Boston, MA: Charles River Media.
- **A2:** Bakie, R. T. (2005). Games and society. In S. Rabin (Ed.), *Introduction to game development* (pp. 37-50). Boston, MA: Charles River Media.
- **A3:** Jarvinen, A. & Holopainen, J. (2005). Ludology for game developers: An academic perspective. In S. Rabin (Ed.), *Introduction to game development* (pp. 51-68). Boston, MA: Charles River Media.
- **A4:** Falstein, N. (2005). Understanding fun: The theory of natural funativity. In S. Rabin (Ed.), *Introduction to game development* (pp. 69-97). Boston, MA: Charles River Media.
- **A5:** Barry, I. (2005). Game design. In S. Rabin (Ed.), *Introduction to game development* (pp. 99-160). Boston, MA: Charles River Media.

Fundamentals of Computer Graphics, P. Shirley, Wellesley, MA, A. K. Peters, 2005.

- **B1:** Sung, K. (2005). Building interactive graphics applications. In P. Shirley (Ed.), *Fundamentals of computer graphics* (pp. 401-449). Wellesley, MA: A K Peters, Ltd.

Programming game AI by example, M. Buckland, Plano TX, Woodward Publishing Inc, 2004.

- **C1:** Buckland, M. (2004). A math and physics primer. In *Programming game AI by example* (pp. 1-41). Plano, TX: Woodward Publishing, Inc.
- **C2:** Buckland, M. (2004). State-driven agent design. In *Programming game AI by example* (pp. 43-83). Plano, TX: Woodward Publishing, Inc.

- Tricks of the Windows game programming guru, Indianapolis, IN, Sams Publishing, 2002.
- **D1:** LaMothe, A. (2002). Playing God: Basic physics modeling. In *Tricks of the Windows game programming gurus* (pp. 803-878). Indianapolis, IN: Sams Publishings.
- Game Design Workshop a play centric approach to creating innovative games, Tracy Fullerton, Burlington, MA, Morgan Kaufmann, 2008.
- **E1:** Chapter 7: Prototyping
- **E2:** Chapter 9: Playtesting
- **E3:** Pages 349-362
- **E4:** Page 394-402

Game Design and Development, E. Adams and A. Rollings, Berkeley, Upper Saddle River, NJ, Pearson Prentice Hall, 2007.

- **F1:** Chapter 4: Game world
- **F2:** Chapter 8: Creating the User Experience
- **F3:** Chapter 9: Game play

Approximated Schedule

Week	Topics	Reading	Date	Notes (subject to changes)
1	Intro + Game Objects	A1-A3, B1	Sep 30	Assign: MP1 + Fav Game
2	My Fav Game (Part-1) + Autonomous Behaviors My Fav Game (Part-2) + Find your teams	C1, C2	Oct 5 Oct 7	Due: MP1 + Fav Game Assign: MP2 + Form your teams
3	Game World: Camera, State, Size Making games fun	D1, F1, F2	Oct 12 Oct 14	Due: MP2 Assign: MP3
4	Brainstorming Tools we use (part-1) + Final Project Expectations + Proposal + Prototypes	F3, AA, EA, A5	Oct 19 Oct 21	Due: MP3 + <u>Finalize team details</u> Assign: MP4 + Final Project
5	<u>Game Objects</u> : Taking notes: Game Objects , Narrative Director: Narrative Director <u>Tools we use</u> (part-2)	E3, E1	Oct 26 Oct 28	Due: MP4
6	Pitch your games Tools we use (part-3)		Nov 2 Nov 4	Due: Game Proposal + GDD + Tools Talk
7	Digital prototype demo Veteran's Day	E2	Nov 9 Nov 11	Due: Prototypes +GDD + Tools Talk
8	Game demo Playtesting and what is to come		Nov 16 Nov 18	Due: Demo +GDD + Tools Talk
9	Alpha playtest Thanksgiving break (alpha playtest feedback due)		Nov 23 Nov 25	Due: Alpha + Feedback for others
10	Beta playtest <u>Game Objects</u> : The other (outside of coding) sides of videogame development: Game Objects (Production), Game Objects		Nov 30 Dec 2	Due: Beta + Feedback for others

11	Retrospective: how is it going, how did it go? GameEngine: Control in game dev Game Engine CDS444N, Technical Director Rui Zuo, Google		Dec 7 Dec 9	Due: Retro talk
12	Final Game Eval		Dec 14	Due: Final game

Game Engine

We will be working with the [Unity game engine](#). Please refer to [this help page](#) to get started on Unity.

Submitting Programming Assignments (MPs)

Submitting Source Code: You will submit your source code of each programming assignment (or machine problem, or mp) and the grader will run/test your submissions. We will be using the canvas facility (refer to the course web-site for the canvas submission link).

- **Source code:** Zip up your entire project, the requirement is that we need to be able to run your submission as a Unity project. Please refer to what I provide you in the course materials, we want your entire project. If we cannot open your project from the Unity Hub, you will lose significant credit for the assignment.
- **Folder name:** Use your first and last name and mp# as the name of your folder and project name. If the name of your game do not tell us who you are, you will lose credit for the assignment.
- Submit as many times as you wish, the grader will only look at the last one received before the deadline. Please **do not** submit hard copies of your program. Let's save some trees, we will look at your source code electronically.

There is a "[Test Assignment Submission](#)" assignment for you to *try things out*. Please do try it before the first MP due date! If I do not see a test submission *and* you have

trouble submitting your first assignment at the last minute, I **will not** help you. I can only help those who tried.

In addition, and **very importantly**, you should always download your submission, un-zip/run/open-in-editor to ensure your submission is correct. Remember, the grader will download your submission, unzip, double-click on the .exe file to run. If the grader cannot download, double click and run your program, you will receive a zero. You have been warned.

You are responsible to ensure that the files you submitted are correct. On the due date of the mp, the corresponding directory will be **close** at precisely the due time. After which, you **will not be able to submit your work!** We **will not** accept submissions via emails. You are responsible to ensure that the files you submitted are correct. Minor submission mistakes (e.g. missed a small file) will result in significant deduction from the assignment. Major submission mistakes (e.g. forgot to include a major source code file) will be treated as incomplete assignment and you will get **0%** for the assignment. On a case-by-case basis, I will decide if a submission mistake is minor or major. There will be no exceptions!

If there is an emergency and/or personal difficulty, please talk to me in person. Remember to document your code, and practice the good programming skills.

General Policies

Assignment Deadlines: There will be no late assignments accepted. Let me put this in another way, there will be no late assignments accepted. This is the case for both homework assignments and machine problems. Pay attention to the deadline on the assignments (including the time), there will be no late assignments accepted. Let me explain this again, there will be no late assignments accepted. I am actually a reasonable person, come talk to me about exceptional circumstances. You know the deadlines now please plan ahead.

Lateness to classes: It does not bother me, just don't disturb anyone. If you want to leave early, it would be very nice if you could give me advance warning. If that's too much trouble, or if you forgot, don't worry, just don't disturb anyone and leave quietly.

Commitments and such: I am usually very easy going. I like relaxed classrooms for learning and will try my best to create such an environment. Please do not confuse relax environment with relax requirements. I work hard, and expect students to work as hard. On average, each percentage of your assignments should represent one-two hours of outside-of-class time. For example, if an MP worth 8%, then on average, you will probably need about 10-15 hours to finish this assignment. Please use this as a reference and let me know if you are spending too much time on the assignments. If most of you are experiencing the same problem, then we will have to adjust the amount of work. Please consider if you have the time this quarter for this class. If you do have the time, please stay in this class, I will work hard and try my best to make this class a worthwhile learning experience for you.

Group Assignments: The final project is a group assignment. You **must** form groups of two (or three if there is a left over person) to work on the final project. No groups can be less than two members.

Problems

If you have any problem with this course, please talk to me as soon as possible. I would like to help in any way I could, but I have to know there is a problem. If you fall behind in this class, it will be difficult to catch up.

Special Needs

If you believe that you have a disability and would like academic accommodations, please contact [Disability Resources for Students](#) (UW1-175) at 425.352.5307 or at drs@uwb.edu. In most cases, you will need to provide documentation of your disability as part of the review process. I will coordinate with the University to ensure that the appropriate accommodations are made in this class.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course. If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact [Disability Resources for Students](#) (UW1-175) at 425.352.5307 or at drs@uwb.edu. DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s), and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

For Our Veterans: If you are a student who has served in our nation's military forces, welcome home, and thank you for your service. I hope that you feel comfortable enough to confidentially self-identify yourself to me so I can help you make a successful transition from the military to higher education.

Religious Accommodation Policy: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](#) (<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#)

[\(https://registrar.washington.edu/students/religious-accommodations-request/\)](https://registrar.washington.edu/students/religious-accommodations-request/).

Academic Conduct

Student Code of Conduct: <http://apps.leg.wa.gov/WAC/default.aspx?cite=478-120>:

"The university is a public institution having special responsibility for providing instruction in higher education, for advancing knowledge through scholarship and research, and for providing related services to the community. As a center of learning, the university also has the obligation to maintain conditions conducive to freedom of inquiry and expression to the maximum degree compatible with the orderly conduct of its functions. For these purposes, the university is governed by the rules, regulations, procedures, policies, and standards of conduct that safeguard its functions and protect the rights and freedoms of all members of the academic community."

...

*"An instructor has the **authority to exclude a student from any class session** in which the student is disorderly or disruptive. If the student persists in the disorderly or disruptive conduct, the instructor should report the matter to the dean of the school or college, or, at the University of Washington Bothell and Tacoma campuses, to the dean or director of the program in which the student is enrolled."*

Academic Integrity and Plagiarism: See

<http://www.uwb.edu/studentservices/academicconduct>

for crucial information regarding academic integrity. The library also has an extremely useful website with resources at <http://libguides.uwb.edu/ai>. You are responsible for knowing what constitutes a violation of the University of Washington Student Code, and you will be held responsible for any such violations whether they were intentional or not. Plagiarism is one of the most common violations of academic integrity, so please pay attention to both the web information and when your instructor explains all of this in class. In short, do your own work, and clearly cite all your sources. If you are unsure, ask for help!

Privacy: The opinion you expressed (in class discussion, in written assignments, on our course discussion board), are yours. None of this information will be shared with anyone, not even your parents.

Other potentially useful/important information

Inclement Weather: Please check if the campus may be closed due to weather. Information about suspension of operations will be made public and available through the media. Students can learn of campus operations status from the website or by calling the Campus Information Hotline 425.352.3333. You may also sign up with an

alert system that will contact you via email or text message if classes are canceled. For more information on the alert process, please see <https://www.uwb.edu/emergency>. Class activities will be rescheduled as needed.

Student Support Services:

IT Helpdesk: IT@uwb.edu , 425-352-3456

Library: <http://library.uwb.edu/> 425-352-5340

Writing Center: www.uwb.edu/WritingCenter/ 425-352-5253

Quantitative Skills Center: <http://www.uwb.edu/qsc> 425-352-3170

Student Success Services: <http://www.uwb.edu/cusp/studentsuccess> 425-352-3776

Career Services: <http://www.uwb.edu/careers> 425-352-3706

Student Counseling Services: <http://www.uwb.edu/student-services/counseling>
425-352-3183